We have the standard properties of a probability function:

- $0 \leq \Pr(\mathcal{A}) \leq 1$ for all measurable sets $\mathcal{A}$.
- $\Pr(\Sigma_{\mathrm{E}}) = 1$.

We also note that lower order objects such as a point or a curve in $\mathbb{R}^2$, or a surface in $\mathbb{R}^3$, have zero measure with respect to Lebesgue measure for the ambient space. Similarly, for a Hamiltonian system with $\mathrm{N_d}$ degrees of freedom, the energy surface is of dimension $2\mathrm{N_d} - 1$ and the probability measure of sets of lower dimension than this will vanish.

As we mentioned in motivating the microcanonical distribution, we could alternatively view microcanonical averages as integrals with respect to the singular measure on the ambient space $\mathcal{D}$ defined by

$$\rho'_{\mathrm{E}} \mathrm{d}\omega = Z_{\mathrm{E}}^{-1} \delta[H - \mathrm{E}] \mathrm{d}\omega,$$

where δ represents the Dirac delta function. The notation $\mathrm{d}\omega$ represents the volume element in the Euclidean space defined by all coordinates and momenta. This is the more common definition of the microcanonical distribution encountered in physics treatments.

Regardless of which formulation is adopted, we may use the probability measure to compute expectations (averages) of (suitable) functions defined on the phase space. We write such an expectation as

$$\mathrm{Av}_{H=\mathrm{E}}(g) = \int\limits_{\Sigma_{\mathrm{E}}} g(\boldsymbol{z}) \mathrm{d}\mu_{\mathrm{E}} = \int\limits_{\mathcal{D}} g(\boldsymbol{z}) \rho'_{\mathrm{E}} \mathrm{d}\omega.$$

Incidentally, any such function is referred to as an *observable function* (or, simply, an *observable*) of the system in this context.

As shorthand, where the underlying space and probability density are assumed, one may write such an average more simply as

$$\mathrm{Av}_{H=\mathrm{E}}(g) = \langle g \rangle.$$

5.4 Temporal Averages and Numerical Computations

Given an autonomous system of differential equations $\dot{\boldsymbol{z}} = \boldsymbol{f}(\boldsymbol{z})$ in $\mathbb{R}^m$, and a trajectory $\Gamma = \{\boldsymbol{z}(t) | t \in [0, \infty]\}$, let $g = g(\boldsymbol{z})$ be any smooth scalar valued function of $\boldsymbol{z}$ (an observable) and define (assuming it exists) the infinite time average of g along Γ by

$$\bar{g}_{\Gamma} = \lim_{\tau \to \infty} \tau^{-1} \int\limits_{0}^{\tau} g(\boldsymbol{z}(t)) \mathrm{d}t. \tag{5.9}$$